



RIVER VALLEY HIGH SCHOOL

YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE NAME

CENTRE NUMBER

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 CLASS INDEX NUMBER

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H2 BIOLOGY

9744/04

Paper 4 Practical

28 Aug 2018

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: As listed in the Confidential Instructions.

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.
 Write your centre number, index number, class and name on all the work you hand in.
 Give details of the practical shift and laboratory, where appropriate, in the boxes provided.
 Write in dark blue or black pen.
 You may use an HB pencil for any diagrams or graphs.
 Do not use staples, paper clips, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
 You will lose marks if you do not show your working, or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

Shift	
Laboratory	
For Examiner's Use	
1	/ 22
2	/ 15
3	/ 13
Total	/ 50

This Question Paper consists of **15** printed pages and **1** blank page.

Answer **all** questions.

- 1 Lipase, **E**, catalyses the hydrolysis of triglycerides into fatty acids and glycerol.

The substrate for **E** will be the triglycerides present in milk, labelled **M**.

The end-point of this hydrolysis can be determined by using an indicator, **I**, which changes colour when the fatty acids are produced.

You are required to:

- prepare different concentrations of the lipase solution, **E**
- investigate the effect of different concentrations of **E** on the hydrolysis of triglycerides in milk.

You are provided with:

labelled	contents	hazard	volume / cm ³
M	milk	none	40
W	distilled water	none	50
I	indicator solution	stains	30
A	solution of alkali	irritant	40
E	5% lipase solution	irritant	50

You are required to dilute the 5% lipase solution, **E**, to provide a range of known concentrations using **simple** dilution.

Decide on the further concentrations of lipase solution you will use in your investigation in addition to the 5% solution.

You will need to prepare 10 cm³ of each lipase solution.

- (a) (i) Prepare the space below to show the concentration of each lipase solution, the volumes of **E** and the volumes of **W**. [3]

- (ii) Describe and explain the expected trend in the time taken to reach the end-point as the concentration of lipase solution increases. [2]

Before starting the investigation, read through steps 1-8 and prepare a table in (a)(iii).

Proceed as follows.

- 1 Prepare **all** the concentrations of lipase solutions you have listed in (a)(i).
- 2 Put 2 cm³ of **M** into each test-tube.
- 3 Put 2 cm³ of **I** into each test-tube containing **M** and gently shake.
- 4 Put 3 cm³ of **A** into each test-tube containing **M** and **I** and gently shake so that all the mixture turns orange. *Note that the mixtures might be different shades of orange.*
- 5 Put 2 cm³ of **E** into one of the test-tubes from step 4 and mix well. Wait for 300.0s. This will be the colour of the end-point.
- 6 Repeat step 5 with **all** concentrations of enzyme solution.
- 7 Start the stopwatch.
- 8 Record the time when each end-point is reached. If the time taken to reach end-point for any one concentration is longer than 300.0 seconds, record as 'more than 300.0'.

- (iii) Record your results in a suitable table in the space below. [3]

- (iv) Calculate the rate of hydrolysis for 5% lipase concentration.
You should show your working and use appropriate units. [2]

- (v) Identify **one** significant source of error in measuring the dependent variable and describe how it affects your results. [2]

- (vi) Other than enzyme concentration, temperature has significant impact on the rate of lipase hydrolysis.

Suggest how you would modify this investigation to obtain an accurate optimum temperature for the activity of E.

[3]

- (b) Some students studied the effect of temperature on the rate of lipase hydrolysis, using a different method that involves determining the presence of triglycerides at various time intervals.

The students' results are shown in Table 1.1.

Table 1.1

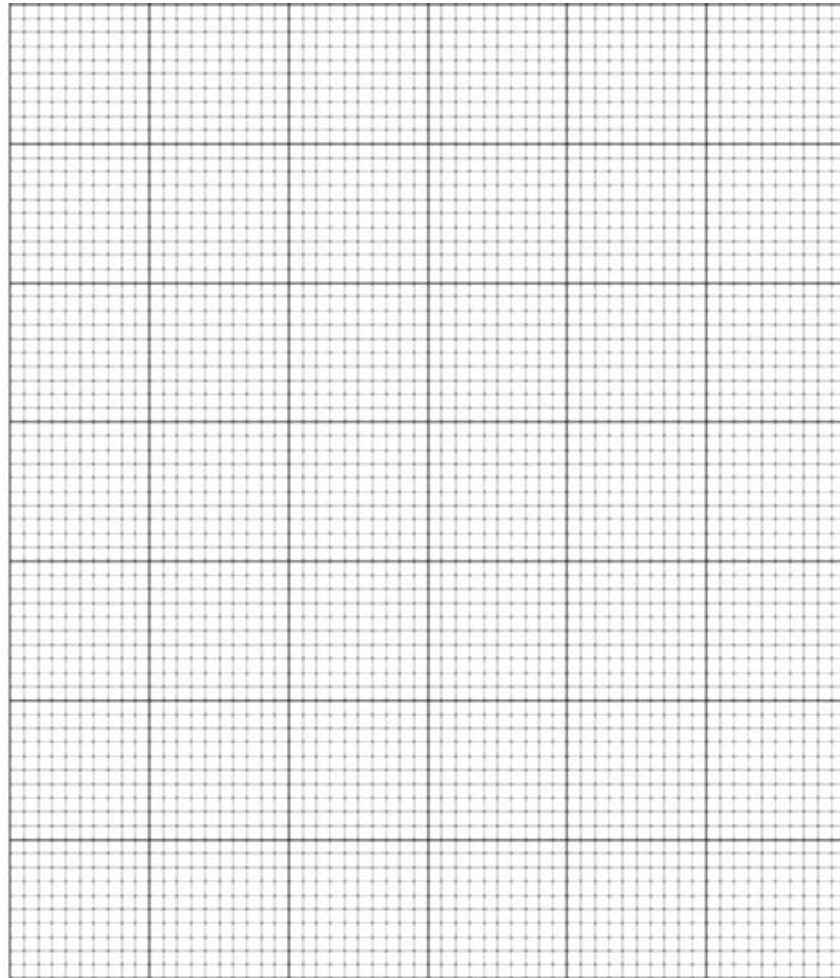
Temperature / °C	Rate of lipase hydrolysis / mol dm ⁻³ s ⁻¹
10	23.0
20	52.0
30	68.5
40	35.0
50	0.5

- (i) Describe a chemical test that can be used to determine the presence of triglyceride in a sample solution.

[2]

- (ii) Use the grid to display the results shown in Table 1.1 in an appropriate form.

[3]



- (iii) Explain the decrease in rate of lipase hydrolysis after 30°C.

[2]

[Total: 22]

- 2 Tuberculosis (TB) is an infectious disease caused by the bacterium *Mycobacterium tuberculosis*. *M. tuberculosis* in the body provokes an immune response, resulting in the production of specific antibodies.

In a test used to detect TB, a modified antibody **A** specific to *M. tuberculosis* is used. Binding of this antibody to *M. tuberculosis* gives rise to an observable result when tested with test reagent **X**.

You are provided with:

- the blood serum of three patients in microfuge tubes labelled **P1**, **P2**, and **P3**
- a suspension of *M. tuberculosis* in a microfuge tube labelled **T**
- distilled water in a microfuge tube labelled **W**
- a solution of the modified antibody in a microfuge tube labelled **A**
- test reagent **X** in a microfuge tube labelled **X**

You are recommended to wear suitable eye protection and gloves. Any splashes on skin should be washed off immediately.

You are required to carry out the test and determine the observations associated with a positive test.

You will then test the blood serums and determine which of the patients should be diagnosed with TB.

You should take care when using the Pasteur pipettes to ensure no cross-contamination of samples and reagents occur.

Proceed as follows.

- 1 Label the wells of the microtiter plate **T**, **W**, **P1**, **P2** and **P3**.
- 2 Add 2 drops of **T** and **W** into the appropriately labelled wells.
- 3 Add two drops of antibody **A** into wells **T** and **W**.
- 4 Add two drops of test reagent **X** into wells **T** and **W**, and leave for 10 seconds.

- (a) Describe your results of testing samples **T** and **W**.

[1]

- 5 Add 2 drops of **P1**, **P2** and **P3** into the appropriately labelled wells.
- 6 Add two drops of antibody **A** into wells **P1**, **P2** and **P3**.
- 7 Add two drops of test reagent **X** into wells **P1**, **P2** and **P3**, and leave for 10 seconds.

(b) Record your results and conclusions in Table 2.1

[2]

Table 2.1

sample	observation	Presence of <i>M. tuberculosis</i>
P1		
P2		
P3		

(c) Explain why antibody **A** cannot be used to detect if a patient is infected with other species of bacteria.

[2]

- (d) Treatment of drug-resistant strains of *M. tuberculosis* requires more than one antibiotic to be used simultaneously. A common treatment involves administering a cocktail solely comprising two component antibiotics, isoniazid and rifampicin.

The relative concentrations of each antibiotic in the cocktail determines its effectiveness. Effective treatment with the cocktail will result in the bacteria being hydrolysed to its constituent biomolecules after 24 hours of exposure.

Table 2.2

cocktail number	relative concentration of isoniazid to rifampicin
1	20-80
2	80-20

A student administered two different cocktails, as shown in Table 2.2, on drug-resistant *M. tuberculosis*, before repeating the test in question 2. He found that both cocktails were ineffective in killing the bacteria.

The student hypothesised that the cocktail is only effective when the relative concentration of each antibiotic component is at least 30%.

Design an experiment to determine the cocktail with the most effective relative concentrations of component antibiotics.

In your plan, you must use:

- a suspension of drug-resistant *M. tuberculosis* in a water-bath at 30°C (3 cm³ of this culture will be required for inoculation of each additional culture)
- a sterile solution of 100% isoniazid
- a sterile solution of 100% rifampicin
- a water-bath at 30°C
- a colourimeter

You may select from the following sterilised apparatus and plan to use appropriate additional apparatus:

- syringes
- 5 cm³ microfuge tubes
- timer, e.g. stopwatch
- a biosafety cabinet
- normal laboratory glassware, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes and pipette fillers, glass rods, etc.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[10]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

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- 3** During this question you will require access to a microscope and slide **S1**.

You are required to use a sharp pencil for drawings.

A blood smear can be used to look for abnormalities in blood cells. Observations made from blood smears allow doctors to diagnose certain blood disorders or other medical conditions.

In a blood smear, mature red blood cells and two categories of white blood cells can be observed clearly. The two categories of white blood cells are lymphocytes and phagocytes.

Mature red blood cells do not have nucleus, but white blood cells contain a nucleus. The shape of lymphocyte's nucleus is round but the shape of phagocyte's nucleus is lobed (dumbbell-shaped, C-shaped etc.).

- (a)** Observe the cells in slide **S1**.

Identify **one** red blood cell and **one** phagocyte.

Use the space provided to draw to the same scale, labelled diagrams of [5]

- (i)** a red blood cell,

- (ii)** a phagocyte.

Slide **S1** is a microscope slide with blood smear of individual **A**.

Fig 3.1 is a photomicrograph of a blood smear of individual **B** viewed at x400.

Both blood smears have been stained using the same technique.

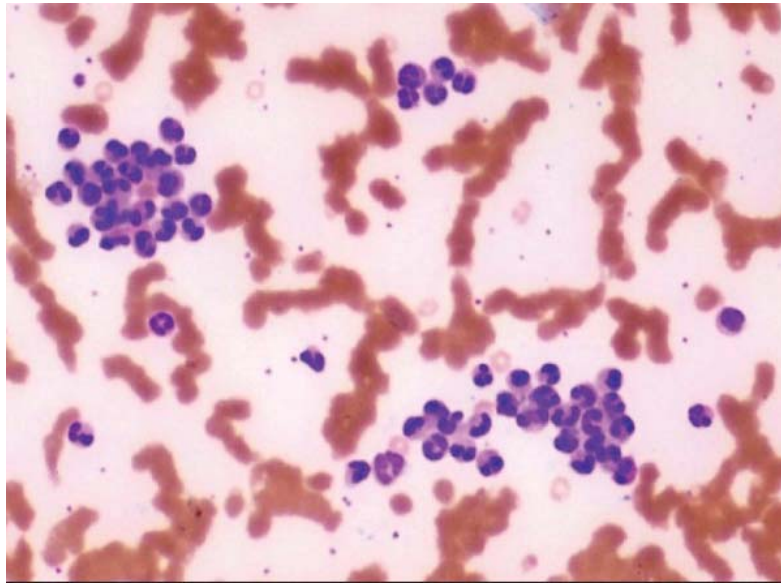


Fig 3.1

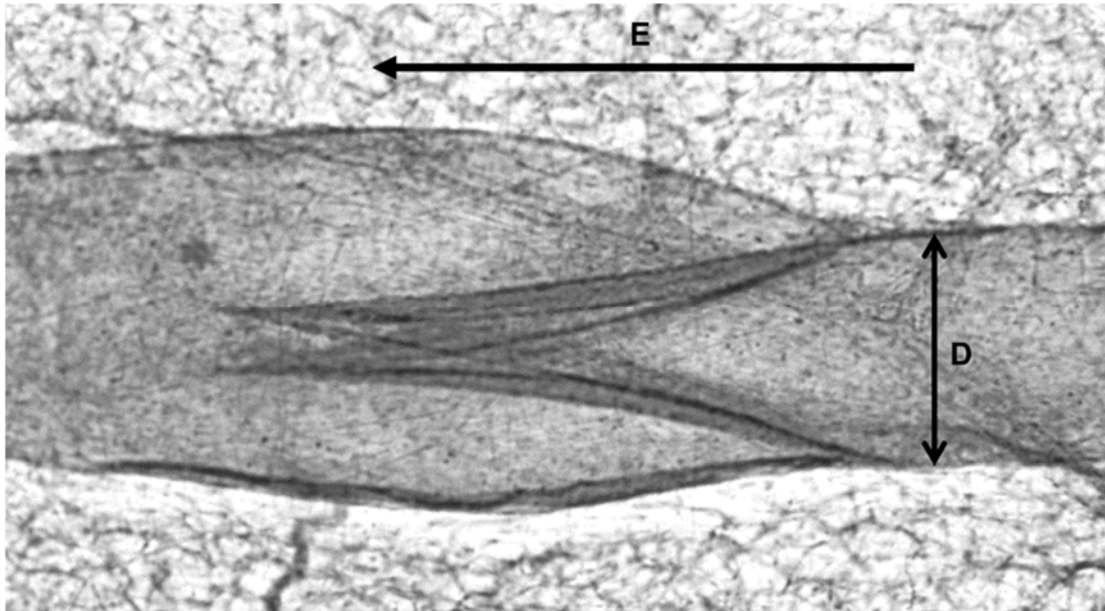
- (b) (i)** Using a suitable form, record observable differences between the blood smear of individual **B** in Fig 3.1 and the individual **A** on slide **S1**. [3]

- (ii)** Individual **B** suffers from pain and shortness of breath. With reference to Fig 3.1, and your own knowledge, suggest reasons for these symptoms. [2]

Lymph is a fluid containing white blood cells, especially lymphocytes, the cells that attack bacteria in the blood. Lymph flows through a network of lymphatic vessels, transporting white blood cells to tissues of the lymphatic system. The lymphatic system includes the bone marrow, thymus and lymph nodes.

Fig 3.2 shows the longitudinal section of a lymphatic vessel with a pair of flap-like structures, known as a valve. Arrow **E** in the photomicrograph shows the direction of lymph flow.

You are not expected to be familiar with this specimen.



X 400.0

Fig 3.2

- (c) Calculate the actual diameter of the narrowest region of the lymph vessel, indicated by line **D**. Show your working clearly. [2]

Actual length of lumen: _____ μm

- (d) With reference to Fig 3.2, describe an observable structural adaptation that allows the lymphatic vessel to transport lymph around the body. [1]

[Total: 13]

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